

3.10 Apply nodal analysis to find node voltages V_1 to V_3 in the circuit of **Fig. P3.10** and then determine I_x .

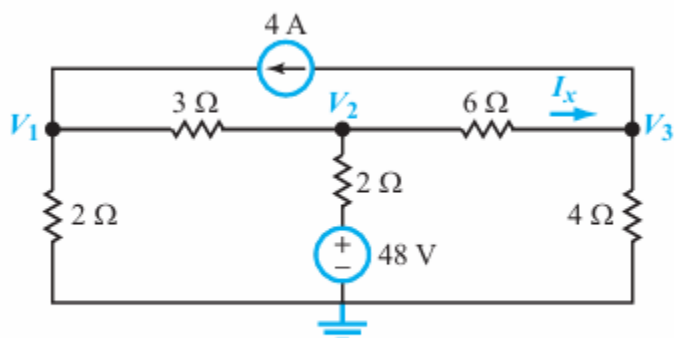


Figure P3.10 Circuit for Problem 3.10.

3.12 The magnitude of the dependent current source in the circuit of **Fig. P3.12** depends on the current I_x flowing through the $10\ \Omega$ resistor. Determine I_x .

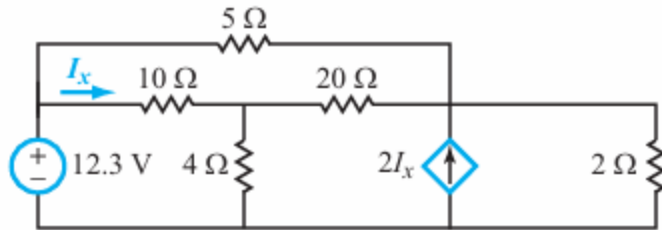


Figure P3.12 Circuit for Problems 3.12 and 3.13.

3.33 Apply mesh analysis to the circuit in **Fig. P3.33** to determine V_x .

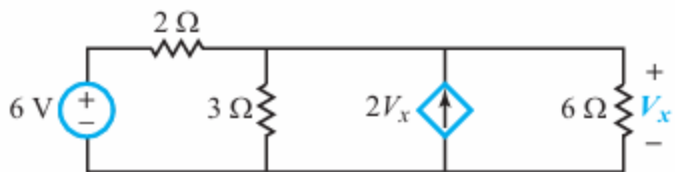


Figure P3.33 Circuit for Problem 3.33.

3.34 Use the supermesh concept to solve for I_x in the circuit of **Fig. P3.34**.

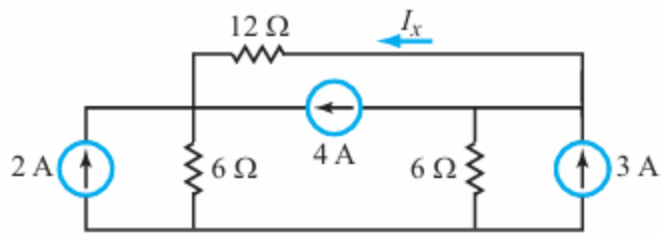


Figure P3.34 Circuit for Problem 3.34.

3.37 The circuit in **Fig. P3.37** includes a dependent current source. Apply mesh analysis to determine I_x .

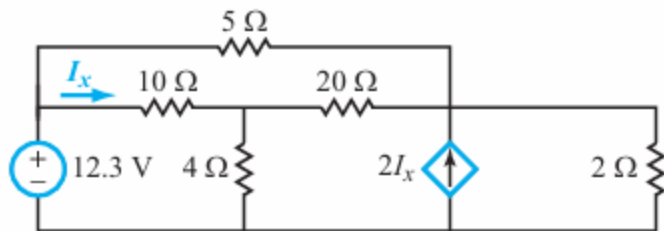


Figure P3.37 Circuit for Problems 3.37 and 3.38.

3.71 Find the Thévenin equivalent circuit at terminals (a,b) of the circuit in **Fig. P3.71**.

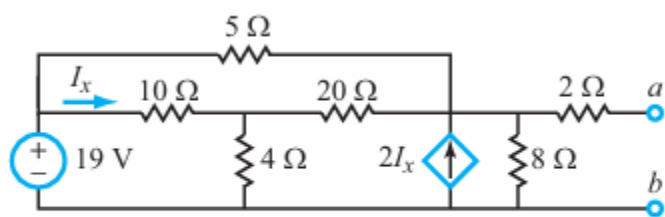


Figure P3.71 Circuit for Problems 3.71 and 3.72.

- *3.74 Find the Norton equivalent circuit at terminals (a, b) of the circuit in Fig. P3.74.

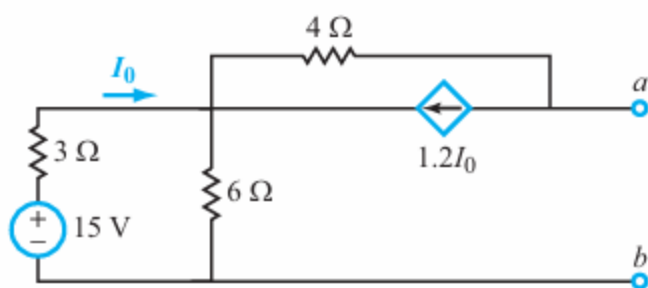


Figure P3.74 Circuit for Problems 3.74 and 3.75.

***4.23** Find the value of v_o in the circuit in **Fig. P4.23**.

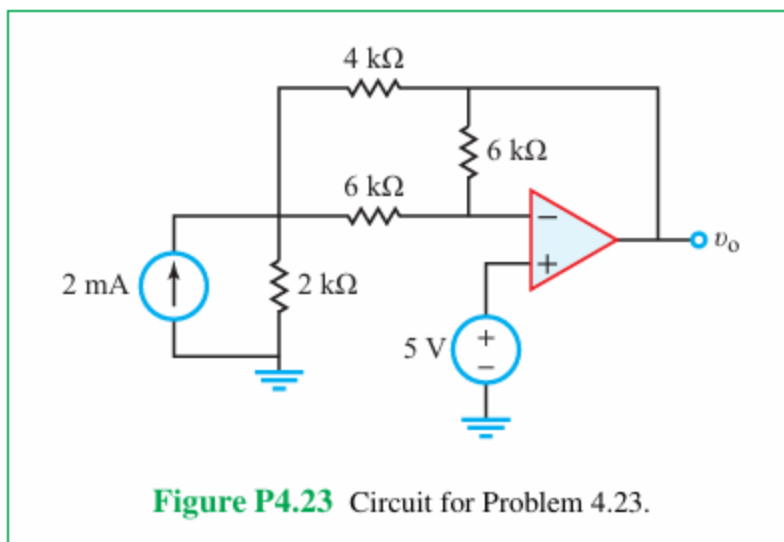


Figure P4.23 Circuit for Problem 4.23.

4.24 For the circuit in **Fig. P4.24**, obtain an expression for voltage gain $G = v_o/v_s$.

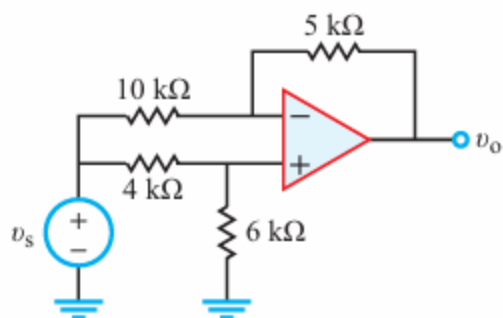


Figure P4.24 Circuit for Problem 4.24.